

# Caching with Amazon DAX – Part 1

## To Begin with the Lab

### Summary of the Lab

In this lab, an **Amazon DAX (DynamoDB Accelerator)** cluster was created and configured to improve **Amazon DynamoDB** read performance using an in-memory cache. A two-node DAX cluster was provisioned with the required **IAM** role, subnet group, VPC, security group, and default parameter group. After the cluster became available, its endpoint was used to configure a Node.js application with the Amazon DAX client. Finally, it was verified that DAX can only be accessed from resources running inside the same VPC, making **AWS Lambda** or **Amazon EC2** suitable environments for testing.

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### Understanding Amazon DAX (DynamoDB Accelerator)

**Amazon DAX** is a fully managed, in-memory cache for **Amazon DynamoDB** that speeds up read operations by serving frequently requested data directly from memory. It requires only minimal code changes because applications simply replace the standard DynamoDB client with the Amazon DAX client. DAX supports only eventually consistent reads, while strongly consistent reads continue to access DynamoDB directly. This improves application performance by reducing read latency from milliseconds to microseconds.

#### Example:

An application stores user profiles in a DynamoDB table named **Notes**. When thousands of users repeatedly request the same profile information, Amazon DAX serves the cached data instead of querying DynamoDB each time, resulting in much faster response times and reduced database load.

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### Lab Steps

#### Step 1: Sign in to AWS

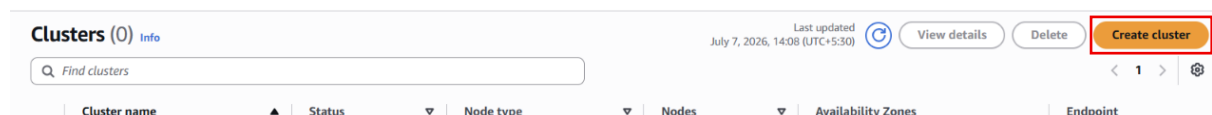
- Sign in to the **AWS Management Console**.

#### Step 2: Open Amazon DynamoDB

- Navigate to **Amazon DynamoDB**.
- Open the **DAX Dashboard**.

#### Step 3: Create a DAX Cluster

- Click **Create Cluster**.



#### Step 4: Configure the Cluster

- Enter the cluster name as **dax-notes-app**.

#### Choose cluster nodes

A screenshot of the 'Cluster name' configuration field in the AWS console. The field is labeled 'Cluster name' and has a description: 'Provide a meaningful name that uniquely identifies your DAX cluster.' The text 'dax-notes-app' is entered into the field and is highlighted with a red rectangular box. Below the field, there's a note: 'Must be between 1 and 20 characters; begin with a letter; contain only ASCII letters, digits, and hyphens; and not end with a hyphen or contain two consecutive hyphens.'

- Select the node type as **dax.r7i.large**
- Set the cluster size to **2 Nodes**.

#### Node families

Choose the type of nodes to run in your clusters. All nodes in each cluster must be of the same type. You cannot modify the node types for a running DAX cluster.

##### Node type family

☒ All families

Compare all node families.

☐ t-type family

Provides a baseline level of CPU performance with the ability to burst above the baseline when needed. Recommended for use cases requiring lower throughput.

☐ r-type family

Each node is allocated with fixed resources, for always-ready capacity.

#### Node types (1/31)

Find node types

< 1 2 3 4 5 6 7 > ⚙

|                                  | Node type       | vCPU | Memory (GiB) | Network performance |
|----------------------------------|-----------------|------|--------------|---------------------|
| <input checked="" type="radio"/> | dax.r7i.large   | 2    | 16.00        | High                |
| <input type="radio"/>            | dax.r7i.xlarge  | 4    | 32.00        | High                |
| <input type="radio"/>            | dax.r7i.2xlarge | 8    | 64.00        | High                |
| <input type="radio"/>            | dax.r7i.4xlarge | 16   | 128.00       | High                |
| <input type="radio"/>            | dax.r7i.8xlarge | 32   | 256.00       | 12.5 Gigabit        |

#### Cluster size

##### Number of nodes

For a cluster requiring high availability, we strongly recommend at least three nodes. You can scale the number of nodes up or down later.

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### Step 5: Configure the IAM Role

- Allow **Amazon DAX** to create a new **IAM** role automatically.
- Enter the role name is **DAXtoDyanamodb**
- Select the **Read/Write** policy.
- Allow access to **all DynamoDB** tables in the Region.

#### IAM permissions

##### IAM Service role for DynamoDB access

Choose the IAM service role that your cluster nodes will assume when accessing your DynamoDB tables. The policy on this role controls which tables they can access, and which operations t

☐ Choose existing

☒ Create new

#### Create new IAM role

##### IAM role name

DAXtoDyanamodb

Use alphanumeric and +,=, @, \_ characters. Maximum 64 characters.

☒ Create policy

Create a new IAM policy and attach it to this IAM role.

##### IAM role policy

Choose policy permissions to attach to this role.

☒ Read/write

☐ Read only

##### New IAM policy name

DAXFullAccess-DAXtoDyanamodb

Use alphanumeric and +,=, @, \_ characters. Maximum 128 characters.

##### Access to DynamoDB tables

Control which DynamoDB tables your DAX cluster nodes would be able to access.

☒ All tables

Allow your DAX cluster nodes to access all your DynamoDB tables.

☐ Specify tables

Limit access only to tables you specify.

### Step 6: Configure the Subnet Group

- Create a new subnet group.

- Enter the subnet group name as **Dax-Subnet-group**.
- Provide a description.
- Select the default **VPC**.
- Choose the required subnets.

### Configure networks

#### Network Settings

**Network Type**  
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

IPv4

**Subnet group**  
DAX will assign network addresses to your cluster nodes from the subnets included in this group. Subnets also determine the Availability Zones (AZs).

☐ Choose existing

☒ Create new

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**New subnet group**

**Subnet group name**  
Dax-Subnet-group

Must be between 1 and 255 characters; begin with a letter; contain only ASCII letters, digits, and hyphens; and not end with a hyphen or contain two consecutive hyphens.

**Subnet group description - optional**  
Dax Subnet group

Maximum 255 characters.

**VPC ID**  
The VPC environment where your DAX cluster will run.

vpc-099792461c7d0a616

 Last updated July 7, 2026, 14:20 (UTC)

### Step 7: Configure Cluster Settings

- Keep the default security group.
- Optionally enter a cluster description.
- Keep the default Availability Zone selection.
- Select the default parameter group.
- Leave the default cache TTL values (300 seconds).
- Leave the maintenance window as the default.
- Keep **Amazon SNS** notifications disabled.

## Parameter group

### Parameter group

Parameter groups are sets of configurations that apply to all the nodes in your cluster.

☒ Choose existing

☐ Create new

default.dax1.0



Last updated  
July 7, 2026, 14:25 (UTC+5:30)

### Details of the chosen group

#### Group description

Default parameter group for dax1.0

[Edit parameter group](#)

#### Item time-to-live (TTL)

5 minutes

#### Query time-to-live (TTL)

5 minutes

## Maintenance window

DAX occasionally applies software upgrades and patches to all of your nodes during a weekly maintenance window. This operation might affect the cluster's performance.

☒ No preference

Maintenance will be done during time windows DAX will determine.

☐ Specify time window

Maintenance will only start during the time window you specify.

## Tags

Tags are pairs of keys and optional values, that you can assign to AWS resources. You can use tags to control access to your resources or track your AWS spending.

No tags are associated with the resource.

[Add new tag](#)

You can add 50 more tags.

[Cancel](#)

[Previous](#)

[Next](#)

## Step 8: Launch the Cluster

- Click **Launch Cluster**.
- Wait until the cluster status changes to **Available**.

### Clusters (1) [Info](#)

Last updated  
July 7, 2026, 17:53 (UTC+5:30)



[View details](#)

[Delete](#)

[Create](#)

| Cluster name                               | Status                 | Node type     | Nodes | Availability Zones | Endpoint                                                                         |
|--------------------------------------------|------------------------|---------------|-------|--------------------|----------------------------------------------------------------------------------|
| <input type="radio"/> <b>dax-notes-app</b> | <span>Available</span> | dax.r7i.large | 2     | 2 zones            | <a href="#">daxs://dax-notes-app.vrl0te.dax-clusters.us-east-1.amazonaws.com</a> |

## Step 9: Review the Cluster

- Open the created cluster.
- Copy the **Cluster Endpoint** for application configuration.

The screenshot shows the AWS Management Console interface for the 'dax-notes-app' cluster. The left sidebar contains navigation links for DynamoDB, Dashboard, Tables, Explore items, PartiQL editor, Backups, Exports to S3, Imports from S3, Integrations, Reserved capacity, and Settings. The main content area shows the cluster details for 'dax-notes-app'. The 'General information' tab is active, displaying the cluster status as 'Creating' and the Amazon Resource Name (ARN) as 'arn:aws:dax:us-east-1:463646775279:cache/dax-notes-app'. The 'Cluster endpoint' is also visible.

## Step 10: Prepare the Application

- Copy the existing Node.js file.
- Rename it to **readOpsDax.js**.

## Step 11: Install the Amazon DAX Client

- Run the following command:
- npm install amazon-dax-client --save**

```

PS C:\dyno> npm install amazon-dax-client --save
npm warn deprecated querystring@0.2.0: The querystring API is considered Legacy. new code should use the URLSearchParams A
PI instead.
npm warn deprecated uuid@3.4.0: uuid@10 and below is no longer supported. For ESM codebases, update to uuid@latest. For
CommonJS codebases, use uuid@11 (but be aware this version will likely be deprecated in 2028).
npm warn deprecated uuid@8.0.0: uuid@10 and below is no longer supported. For ESM codebases, update to uuid@latest. For
CommonJS codebases, use uuid@11 (but be aware this version will likely be deprecated in 2028).
npm warn deprecated aws-sdk@2.1693.0: The AWS SDK for JavaScript (v2) has reached end-of-support, and no longer receives u
pdates. Please migrate your code to use AWS SDK for JavaScript (v3). More info https://a.co/cUPnyil

added 52 packages, and audited 53 packages in 21s

```

## Step 12: Configure the Amazon DAX Client

- Import the Amazon DAX client module.
- Create a new Amazon DAX client object.
- Specify the DAX cluster endpoint.
- Specify the AWS Region.
- Paste this code

```

const AWS = require("aws-sdk");
AWS.config.update({ region: 'us-east-1' });

const AmazonDaxClient = require("amazon-dax-client");
const dynamodb = new AmazonDaxClient({
  endpoints: [],
  region: 'us-east-1'
});

// const dynamodb = new AWS.DynamoDB();
dynamodb.getItem({
  TableName: "td_notes_test",
  Key: {
    user_id: {
      S: "A"
    },
    timestamp: {
      N: "1"
    }
  }
}, (err, data) => {
  if(err) {
    console.log(err);
  } else {
    console.log(data);
  }
});

```

```

JS read-ops-dax.js > dynamodb > region
2  AWS.config.update({ region: 'us-east-1' });
3
4  const AmazonDaxClient = require("amazon-dax-client");
5  const dynamodb = new AmazonDaxClient({
6      endpoints: [''],
7      region: 'us-east-1'
8  });
9
10 // const dynamodb = new AWS.DynamoDB();
11 dynamodb.getItem({
12     TableName: "td_notes_test",
13     Key: {
14         user_id: {
15             S: "A"
16         },
17         timestamp: {
18             N: "1"
19         }
20     }
21 }, (err, data) => {
22     if(err) {
23         console.log(err);
24     } else {
25         console.log(data);
26     }
27 });
28

```

### Step 13: Verify the DAX Deployment

- Confirm that the Amazon DAX cluster is running successfully.
- Verify that the cluster endpoint has been configured correctly.

```

PS C:\Users\rath\OneDrive\Documents\Sandbox> node read-Dax.js
Data: {
  '$metadata': {
    httpStatusCode: 200,
    requestId: 'GE8CE84VD79GEULUGGEC10PQB3JV4KQNSO5AEMVJF66Q9ASUAAJG',
    extendedRequestId: undefined,
    cfId: undefined,
    attempts: 1,
    totalRetryDelay: 0
  }
}
PS C:\Users\rath\OneDrive\Documents\Sandbox>

```